UCC Testbed Hackathon 2026

Participant Readiness Guide

Theme: Innovating Safer Cyber Spaces for Children through Artificial Intelligence

1. Introduction and Event Overview

The UCC Testbed Hackathon 2026 will take place on **5–6 November 2026** at the National Communications Testbed for Emerging and Future Technologies and the Innovation Hub at the Uganda Institute of Information and Communications Technology (UICT), Nakawa, Kampala. This guide provides selected participants with the essential technical, operational, ethical, and logistical information required to prepare before the event and to build, test, and demonstrate their prototypes across the two hackathon days.

Selected participants will be grouped into eight multidisciplinary teams of five members each. The Organisers will form teams and assign each team to a Challenge Track based on complementary skill sets recorded during application, and we will communicate this during the mandatory Pre-Event Online Orientation. Within their assigned track, teams are encouraged to freely explore and select the datasets, pretrained models, frameworks, and open-source tools best suited to their solution. Each team will designate one member as its Team Point of Contact.

Key Readiness Dates and Milestones:

- **30 October 2026 — Pre-Event Online Orientation:** Mandatory virtual orientation session covering the programme, challenge tracks, team allocations, judging rubric, safeguarding rules, SSH key setup, and the participant environment setup guide.
- **5–6 November 2026 — Hackathon Event Days:** Two-day in-person Hackathon at UICT, Nakawa, comprising four structured build sprints, technical mentorship, pitch coaching, code freeze, live final demonstrations, judging, and the awards ceremony.

2. Problem Statement

The rapid acceleration of digital connectivity across Uganda has opened extraordinary avenues for education, creative expression, peer collaboration, and socioeconomic inclusion. However, this unchecked digital frontier has simultaneously precipitated an alarming child protection crisis. Recent empirical research reveals that children and adolescents in Uganda face pervasive, severe online harms that existing oversight systems are failing to arrest.

According to a landmark survey of over 5,700 students conducted by the Uganda Communications Commission (UCC) in Kampala,[1] while 98.4% of surveyed children know how to use the internet, while 88.5% have directly encountered digital harms, including aggressive cyberbullying, hate speech, and online intimidation. Furthermore, 70.1% have been exposed to unsolicited explicit sexual messages, images, or videos. These urban findings mirror national evidence from the joint UNICEF, INTERPOL, and ECPAT Disrupting Harm in Uganda study,[2] which established that 10% of internet-using children aged 15–17 have experienced online sexual coercion, extortion, or unsolicited sexual advances, and 9% had intimate imagery shared without consent or faced extortionate threats of exposure.

Critically, the crisis is severely exacerbated by a systemic "reporting silence": UCC's survey revealed that 74.2% of affected children never report harmful online experiences to anyone including parents, guardians, or teachers.[1] This deep reluctance to report aligns with Disrupting Harm findings, which showed that only 3% to 11% of serious online incidents ever reach formal authorities or child protection social workers, driven by fear of stigma, device confiscation, and the absence of trusted, child-friendly reporting channels.[2]

Compounding these vulnerabilities is the explosive emergence of generative artificial intelligence and deepfake technologies. Global monitoring authorities, such as the Internet Watch Foundation (IWF), recorded a dramatic surge in photorealistic AI-generated child sexual abuse material (CSAM) in 2025—verifying over 8,000 highly realistic synthetic abuse assets, 65% of which depicted the most severe categories of harm.[3] Concurrently, international threat intelligence from the National Center for Missing & Exploited Children (NCMEC) highlights an alarming spike in automated, bot-driven financial sextortion and synthetic identity theft targeting minors.[4]

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[1] Uganda Communications Commission (UCC), Child Online Protection Campaign Survey Findings at Kololo Senior Secondary School, Kampala (October 2026): https://www.ucc.co.ug/ucc-launches-child-online-protection-campaign-at-kololo-secondary-school/

[2] ECPAT International, INTERPOL, and UNICEF, Disrupting Harm in Uganda: Evidence on Online Child Sexual Exploitation and Abuse (2021/2023): https://safeonline.global/wp-content/uploads/2023/12/DH_Uganda_ONLINE_final-Report.pdf

[3] Internet Watch Foundation (IWF), Annual Threat Assessment: AI-Generated Child Sexual Abuse Material and Synthetic Exploitation (2026): https://www.iwf.org.uk/news-media/news/ai-becoming-child-sexual-abuse-machine-adding-to-dangerous-record-levels-of-online-abuse-iwf-warns

[4] National Center for Missing & Exploited Children (NCMEC), CyberTipline Annual Data and Threat Trends in Financial Sextortion (2025/2026): https://www.missingkids.org/cybertiplinedata

Traditional static keyword filters, manual reporting queues, and generic perimeter defenses cannot cope with the sheer volume, velocity, local linguistic subtleties, and sophisticated deception of modern AI-driven threats. The UCC Testbed Hackathon 2026 therefore challenges participants to move beyond superficial mechanisms and build context-aware, scalable AI solutions that proactively identify emerging risks, protect children from exploitation, enable safe and confidential reporting, and foster authentic digital trust for Ugandan children and the broader region.

3. Challenge Tracks

Each team will be assigned to one of the six Challenge Tracks below by the Organisers. Within its assigned track, each team has the freedom to explore the recommended resources below as well as any other suitable open-source datasets, pretrained models, and frameworks.

Track 1: AI for Preventing Cyberbullying and Online Harassment

Develop innovative AI-enabled solutions to identify, prevent or mitigate cyberbullying, harassment, intimidation and other harmful online interactions affecting children. Solutions may address early warning, behavioural patterns, protective interventions or other approaches that promote safer online interactions.

Recommended datasets:

- Civil Comments Dataset - Labelled comments covering toxicity, insults, threats and identity attacks.
- Jigsaw Toxic Comment Classification - Dataset for identifying harmful and abusive online comments.
- Hugging Face Datasets - Additional datasets relating to cyberbullying, harassment and multilingual toxicity.

Recommended pretrained models:

- Toxic-BERT - Toxic language detection.
- Detoxify - Pretrained toxicity classification models.

Track 2: AI for Preventing Online Grooming and Child Exploitation

Explore AI-enabled approaches to recognise, prevent or respond to online grooming, manipulation, coercion and exploitation risks affecting children. Solutions should promote early intervention and child protection while safeguarding privacy and sensitive information.

Recommended datasets:

- PAN12 Sexual Predator Identification Corpus - Historical research dataset for identifying suspicious online interactions. Access and use require safeguarding review.

Recommended pretrained models:

- XLM-RoBERTa - Multilingual language model adaptable to conversational risk analysis.
- Hugging Face Text Classification Models - Models potentially adaptable to suspicious conversation detection.

Track 3: Intelligent Online Content Safety and Age-Appropriate Access

Develop intelligent solutions that help protect children from harmful or age-inappropriate digital content while supporting safe access to educational, social and entertainment resources. Solutions may address content suitability, safer recommendations, adaptive controls or other protective mechanisms.

Recommended datasets:

- Hugging Face Datasets - Content moderation and harmful content datasets.
- TensorFlow Datasets - Image, text and other datasets for developing content assessment models.

Recommended pretrained models:

- Llama Guard Model Family - Safety classification models, subject to applicable access and licence conditions.

- CLIP - Image-text representation model that can support content assessment.

Track 4: AI for Child-Focused Cybersecurity and Online Fraud Prevention

Develop AI-enabled solutions that help protect children against phishing, online scams, impersonation, malicious links, account compromise and other cybersecurity threats. Solutions may focus on threat prevention, detection, awareness or protective interventions.

Recommended datasets and threat intelligence resources:

- PhishTank - Community-maintained phishing information.
- URLhaus - Malicious URL intelligence.
- OpenPhish - Phishing intelligence resources, subject to access conditions.
- UCI Machine Learning Repository - Repository containing cybersecurity and phishing datasets.

Recommended pretrained models:

- Hugging Face Phishing Detection Models - Community-published models for phishing classification.
- Hugging Face Spam Detection Models - Models potentially adaptable to fraudulent message detection.

Track 5: Intelligent Reporting, Response and Referral for Child Online Safety

Develop accessible and secure solutions that improve the reporting, assessment, response and referral of online harms affecting children. Solutions may address incident reporting, prioritisation, case handling, support coordination or access to child protection services.

Recommended datasets:

- Kaggle Datasets – Searchable datasets relating to incident classification and service requests, subject to suitability and privacy review.

Recommended pretrained models and open-source tools:

- Hugging Face Zero-Shot Classification Models - Models potentially useful for incident categorisation and prioritisation.
- Keycloak - Authentication and access control.
- WireMock - Simulated service integrations.

Track 6: AI for Digital Trust and Protection Against Manipulated Content

Explore AI-enabled solutions that help children recognise, assess or respond safely to manipulated media, deepfakes, digital impersonation and misleading AI-generated content. Solutions may promote content authenticity, digital literacy and safer online decision-making.

Recommended datasets:

- FaceForensics++ - Original and manipulated facial video datasets, subject to access approval.
- Kaggle Datasets - Datasets relating to deepfake detection and synthetic media.

Recommended pretrained models and open-source tools:

- Hugging Face Deepfake Detection Models - Community-published deepfake detection models.
- C2PA Rust SDK - Open-source content provenance tools.

4. General Open-Source Resources

Participants may explore the following platforms for additional datasets, pretrained models, development frameworks and implementation examples:

- Hugging Face - Pretrained AI models, datasets and demonstration applications.
- Kaggle - Datasets, notebooks and machine learning examples.
- GitHub - Open-source implementations, frameworks and software components.

- Papers with Code - Research implementations and machine learning benchmarks.
- TensorFlow Hub - Reusable pretrained machine learning models.
- UCI Machine Learning Repository - Machine learning datasets.

NOTE: The datasets, pretrained models, tools and platforms listed under each challenge track are provided for guidance and reference only. They are not mandatory or exhaustive. Participants are encouraged to explore and use other suitable datasets, pretrained models, frameworks, APIs and open-source resources of their choice, provided these are relevant to the selected challenge, comply with applicable licences and data protection requirements, and remain within the scope of the hackathon.

All resources should be assessed for suitability, reliability and safety before use. Particular care is required when handling datasets related to children, online exploitation or other sensitive content.

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5. Technical Environment and Testbed Resources

To support model training, fine-tuning, inference, and integration across all eight teams, the following compute, software, and network resources are provisioned for the Hackathon:

Resource	Provision for Participants	Access and Usage Arrangement
Shared GPU Workstation	On-site Lambda Vector GPU workstation running Ubuntu Linux.	Key-based SSH access over the dedicated event network
Personal Laptops and Testbed Host Workstations	On-site Testbed host workstations available for shared Testbed workloads that require the compute they provide.	Participants are strongly encouraged to bring their own personal laptops and chargers.
Dedicated Event Network and Help Desk	Isolated high-speed event Wi-Fi and internet link covering the main Testbed workspace and the plenary room, supported by an on-site Technical Help Desk.	Network policy and credentials issued at registration.

6. General Rules, Safeguarding and Scope of Solutions

1. **Innovation and Local Relevance:** Participants are free to explore different solutions, technologies, datasets, pretrained models and open-source tools, provided they address the allocated challenge track and apply artificial intelligence or machine learning as a core part of the solution. Solutions are strongly encouraged to consider the Ugandan and

African context, including local languages, accessibility, digital literacy and relevant child online safety challenges.

2. **Prototype Development and Evaluation:** Teams must develop, test and demonstrate a functional prototype or proof of concept within the hackathon period. Every prototype must demonstrate working use of the Testbed infrastructure or provisioned integration services, and detection or classification solutions must define their evaluation method and report performance metrics, including false positives and false negatives.
3. **Data Protection, Safeguarding and Responsible AI:** Solutions and data handling must comply with the Data Protection and Privacy Act, Cap. 97, the Computer Misuse Act, the National Child Policy, the ITU Child Online Protection Guidelines and the UN Convention on the Rights of the Child:
    - Real child sexual abuse material (CSAM), real images of children, and unauthorised personal data relating to identifiable minors are strictly prohibited. Teams may explore and use suitable openly licensed, synthetic, or simulated datasets relevant to their assigned track.
    - Prototypes must be designed to minimise human and child exposure to harmful content. All participants must attend the Safeguarding Clinic on the morning of Day 1.
4. **Third-Party Components, AI Tools and Intellectual Property:**
    - Work on the prototype must be carried out during the Event Days. Teams may build on pre-existing Background IP, open-source software, public datasets, pretrained models, and AI-assisted coding or writing tools, provided every such resource is declared in the submission documentation and used in compliance with its licence terms.
    - Participants retain ownership of their Background IP and retain the right to use, adapt, and further develop the prototype created during the Hackathon in accordance with the Hackathon Participation Agreement, while granting a non-exclusive, royalty-free licence to photograph, record, display, and showcase prototype demonstrations and summaries for promotional, educational, research, and reporting purposes with credit to the team. Mentors and judges have no claim on participant intellectual property.
5. **Deliverables:** By the Code Freeze deadline, each team must submit:
    - A working prototype or proof of concept;
    - Relevant source code and technical artefacts in an accessible repository;

- Brief documentation outlining architecture, setup instructions, evaluation results (including false positives and false negatives), and declarations of third-party assets and AI tools used;
- A three-minute recorded demonstration video; and
- A live final demonstration and pitch highlighting functionality and potential impact.

7. Hackathon Programme and Mentoring

7.1 Programme of the Hackathon

Time	Session	Responsible
Day 1: Thursday, November 5, 2026		
07:30 - 08:30	Arrival, Registration, and Breakfast	Implementing Partner
	Signing of Participation Agreement and Consent Forms	
08:30 - 09:00	Welcome Remarks	1. Mr. Nicholas Mbonimpa, RENU 2. Eng. Rebecca Mayanja, UCC 3. Hon. Nyombi Thembo Dr. Edwin Mugume (Moderator)
09:00 – 09:10	Group Photo	Implementing Partner
09:00 - 09:20	Keynote Speech	Eng. Christine Mugimba
09:20 - 09:50	Safeguarding Clinic	Child Protection Practitioners
10:00 - 14:00	Build Sprint 1	Mentors
14:00 - 17:00	Build Sprint 2	Mentors
17:00 - 17:30	End-of-Day Stand-up	Mentors & Implementing Partner
17:30 - 20:00	Optional Extended Build Session	Implementing Partner
Day 2: Friday November 6, 2026		
07:30 - 08:00	Arrival and Breakfast	Implementing Partner

08:00 – 08:30	Mentor Stand-up & Health Check	Mentors & Implementing Partner
08:30 - 11:30	Build Sprint 3 (Integration Testing)	Mentors
11:30 - 14:00	Build Sprint 4 (Final Demo)	Mentors
12:30 – 14:00	Judges Walk the Floor for Early Context	Judges
14:00	Code Freeze and Technical Verification	Implementing Partner
14:00 - 16:00	Final Demo and Judging	Judges
16:00 – 16:20	Judges Deliberate; Participant Survey	Judges & Implementing Partner
16:20 – 16:35	Stakeholder Remarks	Mr James Atuhaire Tukahirwa, Hiinga Capital
16:35 – 16:50	Award Ceremony	Judges & Implementing Partner
16:50 – 17:00	Closing Remarks	Eng. Christine Mugimba Assoc. Prof. Dorothy K. Okello
17:00 – 17:10	Final Group Photo	Implementing Partner
After 17:10	Departure at Leisure	Implementing Partner

7.2 Mentorship and Pitch Coaching

Five domain mentors will rotate across all eight teams on Day 1 and remain available for consultations on Day 2:

- **Artificial Intelligence and Machine Learning Mentor:** Supports model selection, data pipelines, quantitative evaluation methods, and effective use of the shared GPU workstation.
- **Mobile Networks and Cybersecurity Mentor:** Advises on threat intelligence, filtering and policy design, authentication, and the security posture of prototypes.
- **Product and Commercialisation Coach:** Assists with problem framing, target user identification, business model design, cost estimation, and the route to adoption or procurement.

- **Pitch and Communication Coach:** Coaches teams on narrative structure, live demonstration flow, honest framing of limitations, and time management.
- **Child Protection Practitioner:** Leads the safeguarding Clinic and advises teams on child-centred design, safeguarding compliance, and referral pathways into child protection services.

8. Judging Framework and Awards

8.1 Judging Criteria and Weights

Final demonstrations will be scored by an independent four-member judging panel which will be rotating among the teams to present to them indivually. Every team member present must attend their team's pitch and be prepared to answer questions from the Judges.

#	Criterion	What Judges Look For	Weight
1	**Demonstrated use of the Testbed**	Working use of approved Testbed infrastructure or APIs, demonstrated with test evidence	10%
2	**Technical quality and completeness**	Working functions, appropriate failure handling, readable code and reported evaluation results	20%
3	**Relevance to the theme and the assigned track**	The solution addresses a real child protection problem within the track	15%
4	**Safety by design, privacy and ethical data handling**	Data minimisation, reduction of human exposure to harmful content, and compliance with the content rules	15%
5	**Innovation and originality**	A clear original contribution or useful adaptation, with third-party tools and components identified	10%
6	**Marketability, commercial**	Identified user and market, a credible business model with its costs and revenue, and a	20%

#	Criterion	What Judges Look For	Weight
	viability and real-world impact	route to adoption, commercialisation and scale	
7	**Quality of the demonstration and pitch**	Clear narrative, honest about limitations, delivered within time	10%

8.2 Awards Structure

Position	Awards
Winning team	First prize (cash prize of UGX 3,000,000) and framed individual winner's certificates
First runner-up	Second prize (cash prize of UGX 2,000,000), and individual winners' certificates
All participants	Individual certificates of participation for every participant

Promising prototypes identified during judging will also be recommended to UCC for further development through incubation or other innovation support initiatives.

9. Venue Logistics and Pre-Event Readiness Checklist

9.1 Venue Layout, Catering, Accommodation and Code of Conduct

- **Venue Spaces at UICT, Nakawa:** Registration takes place at the information desk tent outside the Testbed entrance. Development sprints take place inside the Main Testbed Workspace, while plenary sessions and the cool-off/rest area are hosted in the adjacent plenary room (Classroom 13). Overnight stays inside the venue rooms are not permitted; all participants must vacate the workspace by 20:30 at the close of the Day 1 extended session.
- **Meals and Refreshments:** Breakfast, hot buffet lunches, a working dinner on Day 1, and continuous break-room refreshments (coffee, water and soft drinks) are provided free of charge across both days.
- **Transport and Accommodation:** Upcountry participants receive transport reimbursement and three nights of accommodation covering the event window. Participants residing in the Kampala metropolitan area receive a transport allowance and one night of accommodation on Day 1 (5 November 2026) to maintain team continuity between Day 1 and Day 2.
- **Code of Conduct and Security:** Participants must wear their issued event badge and T-shirts visibly at all times on campus, treat all participants, mentors, judges, and venue staff with respect, and use Testbed equipment and the event network strictly for legitimate Hackathon development. Unauthorised software installation, credential sharing outside your assigned team, or malicious network probing is strictly prohibited.

9.2 Participant Pre-Event Readiness Checklist

Before arriving at UICT, Nakawa at 07:30 on Thursday, 5 November 2026, ensure you have completed the following steps:

- Verify your accommodation arrangements ahead of the event days. For more information, please reach out to **Mr. Benjamin Makanga (0784018481)** or **Mr. Onyait Richard (0773101651).**
- Prepared your personal laptop and charger with the required development environment.
- Explored the recommended open-source datasets, pretrained models, and platforms for your assigned Challenge Track to identify the resources your team wishes to use. A WhatsApp community will be created, and you will be added to the respective assigned group pre-event.
- Pack your identification document (National ID, Passport, or University ID matching your application), personal laptop and charger, and arrive ready to sign the Hackathon Participation Agreement at registration.